

Xiaotian Hu

+86 15716592102 | [Email](#) | [Homepage](#)

BIO

A strongly self-motivated third-year undergraduate student majoring in Biomedical Engineering at Beihang University, currently ranked first in the major. Admitted under Beihang's broad Aerospace Engineering category, I ranked 1/416 in my first year and subsequently chose BME as my major. I am a two-time recipient of the National Scholarship and have earned multiple distinctions in academic and research competitions. My research interests include AI agents, multimodal vision-language models, and large language models.

ACADEMIC BACKGROUND

- **Beihang University (BUAA)** Sep 2023 – Expected Jun 2027
Bachelor of Engineering in Biomedical Engineering Beijing, China
- **Academic Performance:**
 - * **Biomedical Engineering** (Sep 2023 – Feb 2026): Average Score: 94.04/100; Rank: 1/50
 - * **General Engineering (Aerospace Track)** (Sep 2023 – Jun 2024): Average Score: 93.54/100; Rank: 1/416
- **Awards:** Two-time National Scholarship (top 0.2%), Merit Student Award, and Outstanding Student Award.
- **Selected Coursework:**
 - * Linear Algebra (100), Numerical Analysis (100), Engineering Graphics (100), Fundamental Physics B (100)
 - * Mathematical Analysis for Engineering (98), Probability and Statistics (97), Methods of Mathematical Physics (96)
- **Technical Skills:** Python, C, PyTorch, MATLAB, SolidWorks, and $\LaTeX$.
- **English:** CET-6: 537; CET-4: 529.

PUBLICATIONS

Published / Accepted

1. **X. Hu**, J. Huang, M. Liu, et al. FetalAgents: A multi-agent system for fetal ultrasound image and video analysis. *International Conference on Medical Image Computing and Computer Assisted Intervention*. (MICCAI 2026 Early Accept, Top 9%).[\[Paper\]](#)[\[Code\]](#)
2. **X. Hu**, et al. Benchmarking spatiotemporal fetal brain atlas construction. *Organization for Human Brain Mapping Annual Meeting*. (OHBM 2026 Poster).
3. **X. Hu**, et al. INSTA: Implicit neural spatio-temporal atlas from thick-slice clinical fetal brain MRI. *ISMRM Workshop on Unlocking the Potential of Prenatal MRI: Advances in Fetal Brain, Heart & Placenta Imaging*. (ISMRM 2026 Oral).
4. Z. Gao, et al., **X. Hu**, and Q. Tian. View-Conditioned Semi-Supervised Learning for Fetal Cardiac Ultrasound Analysis. *IEEE 23rd International Symposium on Biomedical Imaging*. (ISBI 2026).[\[Paper\]](#)
5. Y. Li, M. Liu, L. Chang, et al., **X. Hu**, et al. Maximizing Domain Generalization in Automated Fetal Brain Biometry. *International Conference on Medical Image Computing and Computer Assisted Intervention*. (MICCAI 2026).[\[Code\]](#)

Under Review / Submitted

1. **X. Hu**, M. Liu, J. Huang, et al. Towards reliable fetal ultrasound interpretation with multi-agent collaboration. *Medical Image Analysis*. (MedIA, Major Revision).[\[Paper\]](#)[\[Code\]](#)
2. **X. Hu**, M. Liu, H. Yang, et al. INFANiTE: Implicit neural representation for high-resolution fetal brain spatio-temporal atlas learning from clinical thick-slice MRI. *Proceedings of the AAAI Conference on Artificial Intelligence*. (AAAI 2027, Under Review).[\[Paper\]](#)[\[Code\]](#)
3. H. Yang, M. Liu, Y. Hao, **X. Hu**, et al. Simple Baselines for Fetal Brain Unsupervised Anomaly Detection. *IEEE International Conference on Bioinformatics and Biomedicine*. (BIBM 2026, Under Review).[\[Code\]](#)
4. Y. Luo, Y. Li, M. Liu, et al., **X. Hu**, et al. FetAngle: Toward Generalizable Automated Fetal Brain Angle Biometry via Test-Time Adaptation. *British Machine Vision Conference*. (BMVC 2026, Under Review).[\[Code\]](#)
5. V. Galinova, M. Seifi, et al., **X. Hu**, et al. AI4Life Open Calls and Public Challenges: Why, How, and What We Have Learned. *npj Imaging*. (Submitted).

RESEARCH EXPERIENCE

- **Research Assistant – BIRTH Lab, Tsinghua Biomedical Engineering** *Advisor: Qiyuan Tian*
 - Fetal VQA: Built the first dedicated fetal-ultrasound visual question answering benchmark, together with a multi-dataset evaluation suite spanning seven clinical tasks.
 - Developed *FetalAgents* & *FetUSAgents*, a tool-augmented multi-agent framework for fetal ultrasound visual question answering, report generation, image captioning, and video summarization.
 - **MICCAI Early Accept:** *FetalAgents: A Multi-Agent System for Fetal Ultrasound Image and Video Analysis*.
 - **MIA Major Revision:** *Towards Reliable Fetal Ultrasound Interpretation with Multi-Agent Collaboration*.
 - **INFANiTE:** Proposed an implicit-neural-representation framework for constructing high-resolution spatio-temporal fetal brain atlases directly from clinical thick-slice MRI, reducing the pipeline from days to hours.
 - **AAAI Under Review:** *INFANiTE: Implicit Neural Representation for High-Resolution Fetal Brain Spatio-Temporal Atlas Learning from Clinical Thick-Slice MRI*.
- **Algorithm Engineer – FacePhys & HCI Lab, Tsinghua Computer Science** *Mentor: Jiankai Tang*
 - **Human Foundation Model:** Contributed to a human-centric multimodal streaming foundation model supporting real-time full-duplex interaction.
 - Built data collection pipelines to collect facial videos and livestreams from YouTube and Twitch for model training.
 - Investigated algorithms for blood-pressure estimation from facial videos.
- **Visiting Student – Electrical and Computer Engineering, HKU** *Advisor: Cheng Chen*
 - Validated a medical vision-language model with Euclidean, hyperbolic, and spherical representations, demonstrating strong downstream performance.
 - **NMI Manuscript in Preparation:** Second-author work targeting *Nature Machine Intelligence*.

COMPETITIONS AND CHALLENGES

- **Biomedical Engineering Conference and Innovation Medical Summit** *Poster Presenter*
China *Apr 2026*
 - Poster accepted for presentation at the national conference, organized by the Chinese Society of Biomedical Engineering.
- **AI4Life Supervised Microscopy Image Denoising Challenge** *Participant & Co-author*
International *Sep 2025*
 - Achieved **Second Place** in the AI4Life Supervised Microscopy Image Denoising Challenge.
 - **Co-author:** *AI4Life Open Calls and Public Challenges: Why, How, and What We Have Learned*, submitted to *npj Imaging*.
- **16th Chinese Mathematics Competition for College Students** *Participant*
China *Dec 2024*
 - Awarded **First Prize**.
- **40th National College Physics Competition (Selected Regions)** *Participant*
China *Dec 2024*
 - Awarded **Third Prize**.
- **Mathematical Contest in Modeling** *Participant*
International *May 2024*
 - Awarded **Meritorious Winner**.